99/100

CHE 313 Transport Phenomena III

Exam #3 (11/20/2025)

target Comp: n-decane

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1. (25pts) A simple steam distillation is being done in the apparatus. The organic feed is 85 mol% n-decane and 15% nonvolatile organics. The system is operated with liquid water in the still. A total condenser is used to produce $\boxed{\text{pure n-decane}}$ from distillate (water+n-decane). Distillation is continued until the organic layer in the still is 5 mol% n-decane. F = 10 kmole. Pressure is 760 mm Hg.

   ✓ 1)(5pts) Draw a schematic of the steam distillation. Set up overall mass balance and component mass balance equations

   ✓ 2)(5pts) Derive the equation for W_f (amount liquid remaining in the pot)

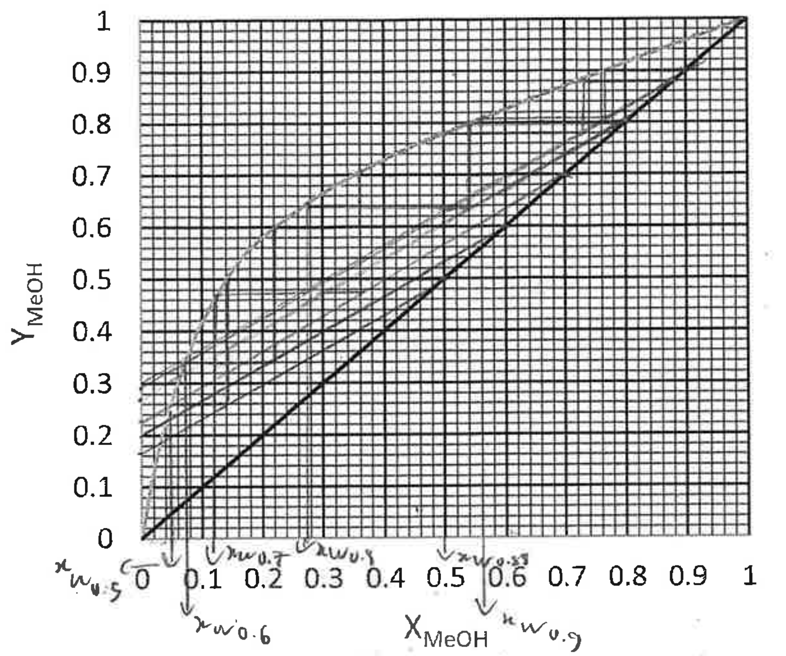
   ✓ 3)(5pts) Calculate W_f and D_{total}

   4)(10pts) Calculate W_f and D_{total} using the Rayleigh equation for simple batch distillation. Compare your answer with one from (3). (Hint: the produced n-decane is pure). ✓

$$ln\left(\frac{W_f}{F}\right) = -\int_{x_{w,final}}^{x_{w,initial}} \frac{dx_w}{x_D - x_w}$$

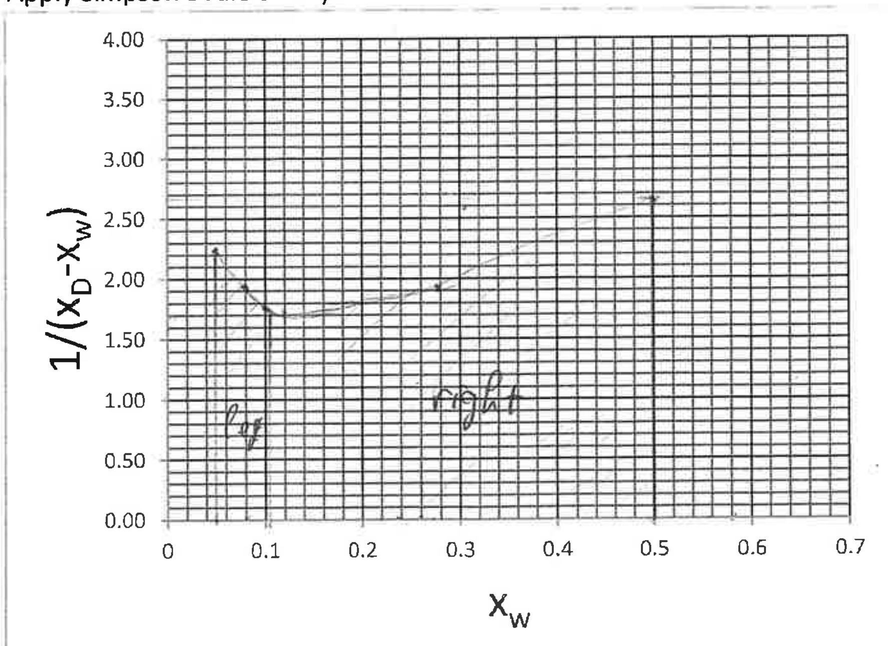
2. (40 pts) 4.0 kmoles of a feed that is ~~60~~ 50 mole % methanol and 40 mole % water is batch distilled in a system with a still pot (an equilibrium contact), a column that acts as one equilibrium stage (total two equilibrium contacts), a total condenser, and reflux to the column. p = 1.0 atm, the reflux ratio is L/D = 2.0, CMO is valid, and $x_{W,final}$= 0.08.
   1) (5pts) Draw a schematic of the batch distillation. Define all parameters including F, W_f, V, L, and D_{Tot}. Set up overall mass balance and component mass balance. ✓
   2) (5pts) Set up equation for $x_{D,\,avg}$ from component mass balance equation. ✓
   3) (5pts) Find the slope of operation equation, y = (L/V)x + (1-L/V)x_D ✓
   4) (10pts) Complete the table below using the VLE graph (draw OP lines and mark your x_w values on the graph).

y-int.	x_D	x_W	$x_D - x_W$	$1/(x_D - x_W)$
0.293	0.88	0.5	0.38	2.63
0.267	0.8	0.275	0.525	1.90
0.233	0.70	0.12	0.58	1.72
0.2	0.60	0.08	0.52	1.92
0.167	0.50	0.05	0.45	2.22



(5pts) Build the integration curve for x_w vs. $1/(x_D - x_w)$ and calculate the area under the curve (Hint: Apply Simpson's rule twice)

2 regions



5) (10pts) Find W_{final}, D_{total}, and $X_{D,avg}$.

Simpson's rule:

$$Area = \frac{x_{Feed} - x_{final}}{6}\left[\frac{1}{x_D - x_w}\bigg|_{x_{final}} + \frac{4}{x_D - x_w}\bigg|_{x_{w,middle}} + \frac{1}{x_D - x_w}\bigg|_{x_{Feed}}\right]$$

Rayleigh equation: $W_{final} = F\, exp\left(-\int_{x_{final}}^{x_{Feed}} \frac{dx_w}{(x_D - x_w)}\right)$

3. **(35 pts) We are absorbing acetone at 25C into water. 1000.0 kmol/hr of air containing 100 ppm of acetone and 2.0 atm is to be processed. Entering water from stripper already contains 0.1 ppm of acetone. Recovery of acetone in the water is 80%. At 25 °C, equilibrium relation is y = 105.6x.**

   a) (5 pts) Draw a schematic diagram of the distillation column and label all known variables. Set up overall mass balance and component mass balance around stage j.

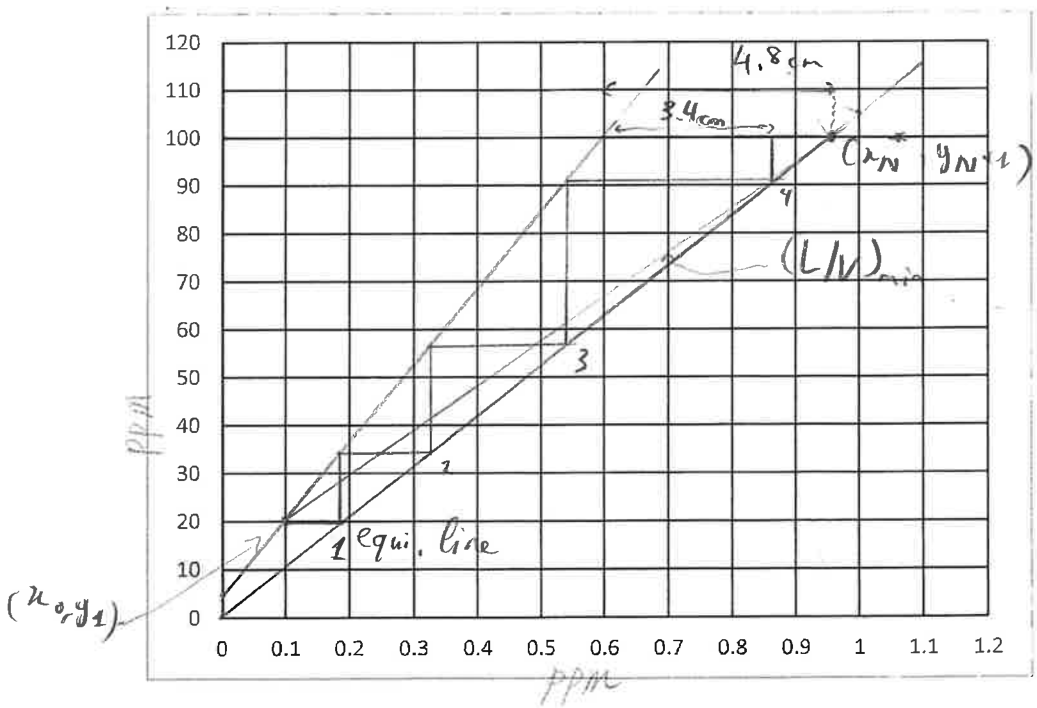
   b) (5 pts) Derive operating equation using component mass balance.

   c) (5 pts) Find (x_0, y_1) and (x_N^*, y_{N+1}). x_N^* is x in equilibrium with y_{N+1}.

   d) (5 pts) Plot equilibrium and minimum operating lines.

   e) (5 pts) When L/V = 1.59 $(L/V)_{min}$, find L/V, x_N, and y-intercept of operating equation

   f) (10 pts) Plot operating line. Find the number of equilibrium stages.

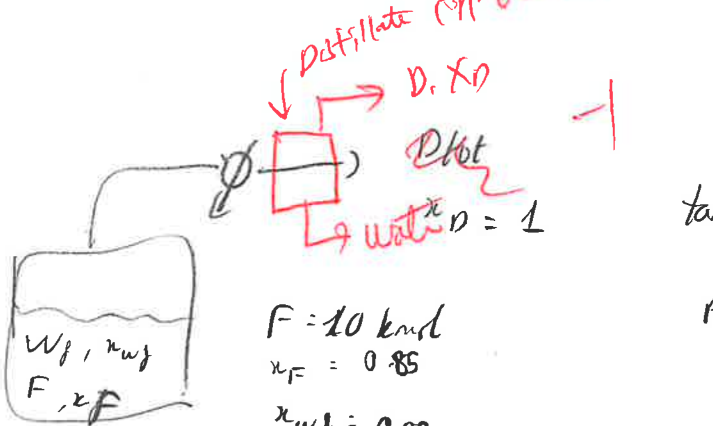


Quan Huynh

①

Q.1:

1)



$F = 10$ kmol
$x_F = 0.85$
$x_{wf} = 0.05$

target comp: n-decane

$P = 760$ mmHg $= 1$ atm

4/5

$$M.B: \begin{cases} F = D_{tot} + W_f \\ F x_F = D_{tot}\, x_D + W_f\, x_{wf} \end{cases}$$

2) $$\boxed{W_f = \frac{1}{x_{wf}}\left(F x_F - D_{tot}\, x_D\right)}$$

4/5

OK. But → $W_f = F \cdot \left(\frac{X_D - X_F}{X_D - X_f}\right)$

3) From M.B:

$$\begin{cases} 10 = D_{tot} + W_f \\ 10 \cdot 0.85 = D_{tot} + 0.05 W_f \end{cases} \Rightarrow \begin{cases} D_{tot} = 10 - W_f \\ D_{tot} = 8.5 - 0.05 W_f \end{cases}$$

5/5

$$\Rightarrow 10 - W_f = 8.5 - 0.05 W_f \Rightarrow \boxed{W_f = 1.58 \text{ kmol}}$$

$$\Rightarrow D_{tot} = 10 - 1.58 = \boxed{8.42 \text{ kmol}}$$

4) $$\ln\left(\frac{W_f}{F}\right) = -\int_{x_{wf}}^{x_{wi}} \frac{dx_w}{x_D - x_w} = -\int_{x_{wf}}^{x_F} \frac{dx_w}{x_D - x_w} = -\int_{0.05}^{0.85} \frac{dx_w}{1 - x_w}$$

$$= \ln(1 - x_w)\Big|_{0.05}^{0.85} = \ln\left(\frac{1 - 0.85}{1 - 0.05}\right) = -1.846 = \ln\left(\frac{W_f}{F}\right)$$

$$\Rightarrow W_f = F e^{-1.846} = 10\, e^{-1.846} = \boxed{1.58 \text{ kmol}}$$

10/10

, $D_{tot} = 8.42$ kmol

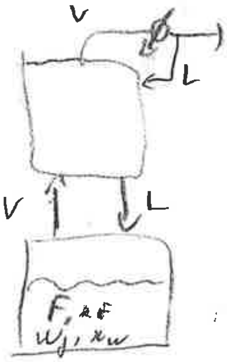
⇒ Compared to part 3), result of part 4) is very similar, if not the same

(2)

Q.2

1)

Target comp: methanol



D_{tot}

x_{Davg}

$F = 4$ kmol

$x_F = 0.5$

$x_{wf} = 0.08$

$L/D = 2 \Rightarrow$ fixed L/D, varying x_D

$P = 1$ atm

1) 5/5

M.B: $\begin{cases} F = D_{tot} + W_f \\ F x_F = D_{tot}\, x_{Davg} + W_f\, x_{wf} \end{cases}$

2) $x_{Davg} = \frac{1}{D_{tot}} \left(F x_F - W_f\, x_{wf} \right)$

2) 4/5

$X_{D,avg} = \frac{F \cdot X_F - W_F \cdot X_f}{D_{tot}}$

3) Top column M.B: $V = D + L$

$\Rightarrow \frac{L}{V} = \frac{L}{D+L} = \frac{L/D}{1+L/D} = \frac{2}{2+1} = 2/3$

$\Rightarrow$ slope for op. line $y = L/V\, x + (1 - L/V) x_D$ is $\boxed{L/V = 2/3}$

3) 5/5

4) 10/10

5) Since the area can be better approxed by dividing the whole area into 2 segments, 0.05 to 0.12 and 0.12 to 0.5

Mid point for $0.05 \to 0.12$: $\frac{0.12 + 0.05}{2} = 0.085 \approx 0.08$ for convenience.

5) 5/5

mid point for $0.12 \to 0.5$: $\frac{0.12 + 0.5}{2} = 0.31$, we use 0.275 for convenience.

6) 10/10

$\Rightarrow A_{tot} = A_{left} + A_{right} = \left[\frac{0.12 - 0.05}{6}\left(2.22 + 4 \times 1.92 + 1.72\right)\right] + \left[\frac{0.5 - 0.12}{6}\right]\left(2.63 + 4 \times 1.90 + 1.72\right)$

$= 0.8924$

6)

$\Rightarrow W_f = F\, e^{-0.8924} = 4\, e^{-0.8924} = \boxed{1.64 \text{ kmol}}$

$D_{tot} = F - W_f = 4 - 1.64 = \boxed{2.36 \text{ kmol}}$

Continue in box above.

5) (cont)

$x_{Davg} = \frac{1}{D_{tot}}\left(F x_F - W_f\, x_{wf}\right)$

$= \frac{1}{2.36}\left(4 \times 0.5 - 1.64 \times 0.08\right)$

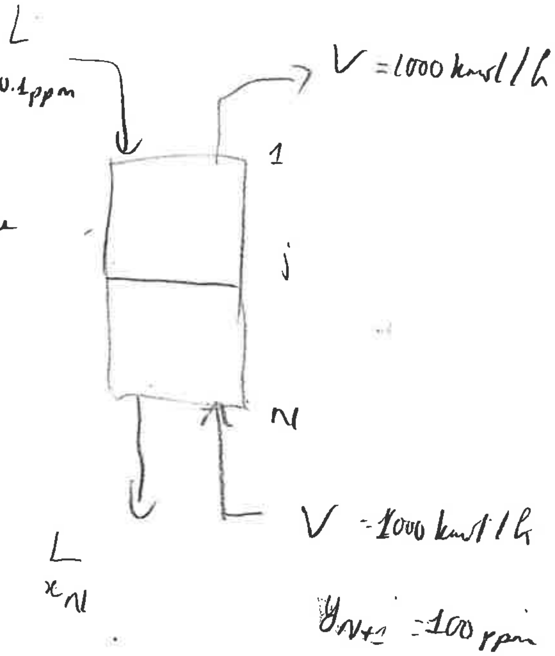
$= \boxed{0.79}$

Quan Huynh

3

a)

Target comp: acetone



equi line: $y = 105.6x$

recovery in water: 80%

assume CMO condition..

a) 5/5

b) 5/5

Overall MB:

$$\begin{cases} L_0 + V_{N+1} = V_1 + L_N & L + V = V + L \quad \text{(CMO condition)} \\ L x_0 + V y_{N+1} = V y_1 + L x_N \end{cases}$$

MB around stage j:

$$L x_0 + V y_{j+1} = V y_1 + L x_j$$

b) $$y_{j+1} = \frac{L}{V} x_j + \left(y_1 - \frac{L}{V} x_0\right)$$

c) 5/5

d) 5/5

e) 5/5

f) 12/10

(+2) Fraction calculation!

c) 80% recovery of acetone in water => 20% remain in gas stream

=> $y_1 = 100 ppm \times 20\% = 20 ppm$

x_N^* is in equilibrium with y_{N+1}

=> $y_{N+1} = 105.6\, x_N^*$ => $x_N = \frac{100}{105.6} = 0.947\ ppm$

$(x_0, y_1) = (0.1, 20)$

$(x_N^*, y_{N+1}) = (0.947, 100)$

e) (4)

$$(L/V)_{min} = \frac{y_{N+1} - y_1}{x_N^* - x_0} = \frac{100 - 20}{0.947 - 0.1} = 94.45$$

$$\Rightarrow \frac{L}{V} = 1.59 (L/V)_{min} = 1.59 \times 94.45 = \boxed{150.2}$$

For stage N:

$$y_{N+1} = \frac{L}{V} x_N + \left(y_1 - \frac{L}{V} x_0\right)$$

$$\Rightarrow 100 = 150.2\, x_N + (20 - 150.2 \times 0.1)$$

$$\Rightarrow x_N = \boxed{0.633}$$

$$y\text{-int} = y_1 - \frac{L}{V} x_0 = 20 - 150.2 \times 0.1 = 4.98$$

f) After 4 equilibrium stages

residual fraction $\frac{3.4}{4.8} = 0.70$

$\Rightarrow$ Total equilibrium stages: 4.7 stages